

INTRODUCTION

Principal frontend engineer with 20+ years of experience architecting extensible UI platforms and enterprise-scale SaaS applications. Expert in TypeScript/React, browser performance, and UI system design, with strong debugging instincts and a track record of improving reliability through testing, tooling, and CI/CD hygiene. Effective cross-functional partner to Product, UX, and Backend teams—shaping API contracts and implementation patterns that make complex workflows feel simple. Thrives in high-ownership environments where speed matters, quality is non-negotiable, and the platform must scale securely.

EDUCATION

CARNEGIE MELLON UNIVERSITY, Pittsburgh, PA

Bachelor of Science, Electrical and Computer Engineering, 2007

SKILLS & PROFICIENCIES

- Technologies JavaScript, TypeScript, CSS/Sass, SSR, Browser APIs, iframes, PostgreSQL, AWS Lambda, DynamoDB, C#
- Tools React, Vue, Node, RTK, MUI, Tailwind, d3, GSAP, Three.js, Pixi, Jest, Cypress, Vitest, Vite, Rollup, Webpack
- AI Copilot, Claude Code, MCP, prompt engineering, AI-assisted development, testing, and content creation
- Apps VSCode, Miro, Jira, Linear, Confluence, Figma, XD
- Deliverables Multi-tenant SaaS platforms, UI frameworks, data visualizations, desktop apps, games, and documentation

PROFESSIONAL EXPERIENCE

Biolumina, New York City, NY

Freelance Software Development Engineer, 03/2025–Present; Remote

- Thrived in a fast-paced healthcare agency environment with shifting scope, tight deadlines, and regulated review cycles
- Built reusable UI frameworks across Vue/React/vanilla JS to accelerate delivery for teams and AI-assisted workflows.
- Built a dedicated Node.js/Express middle-layer (Electron-wrapped) to act as a backend-for-frontend for the UI, simplifying releases and standardizing how the app integrated with downstream services and assets.

Metabolon, Research Triangle Park, NC

Principal Software Development Engineer, 08/2017–02/2025; Remote

- Architected and evolved the company's primary React web app from greenfield to a production, enterprise-scale, multi-tenant SaaS platform serving all client organizations for data delivery and analysis.
- Developed and maintained a shared UI framework to enforce consistency and accelerate delivery across three engineering teams; reduced duplicated UI implementation effort by ~30%
- Drove UI performance and reliability improvements through deep debugging with Chrome DevTools, Lighthouse, and AI-assisted workflows (profiling, tracing, and audit-driven fixes), preventing recurring production issues.
- Designed a comprehensive testing strategy using Jest and Cypress; established E2E and unit test coverage standards and patterns that improved release confidence and reduced regressions across the platform
- Authored technical documentation, UML diagrams, and optimization roadmaps; identified and prioritized technical debt
- Built rich data visualizations with web workers and virtualization to process large data sets without blocking the main thread
- Utilized browser-native storage APIs (IndexedDB via Dexie) to improve performance and support offline-capable workflows
- Owned and evolved the front-end build pipeline (CRA, Webpack, Vite) and drove modernization: led adoption of hooks and the systematic refactoring of legacy class components to functional components, moved from Thunk to RTK Query
- Maintained CI/CD pipelines covering security vulnerabilities, automated builds, testing, and deployment
- Shipped high-quality front-end features across analytics and workflow in close collaboration with product and design
- Monitored production performance and error trends post-release; iterated based on telemetry and user feedback.
- Partnered with Product and UX to translate complex requirements into UI solutions that balance usability with performance
- Integrated AI tools (GitHub Copilot, Claude) into workflows to accelerate testing, code reviews, and component generation
- Led internal training sessions on AI-assisted tooling, improving developer experience and overall delivery throughput
- Interviewed and onboarded engineers; distributed tasks, reviewed code, and provided feedback on patterns and style

NeON, an FCB Health Company, New York, NY*Freelance Senior Developer, 11/2013–05/2017; Remote*

- Architected and delivered responsive websites using Bootstrap, Handlebars, Angular, React, Vue, Next.js, and Node.js
- Built Node.js/Electron-based desktop servers supporting multi-machine, networked application rollouts at live conferences
- Authored mkr.js, an open-source JavaScript framework abstracting HTML/CSS authoring — used across the team to accelerate templated content production
- Developed web and interactive content for CRM tools, games, banner advertisements, and convention panels
- Designed a robust analytics platform for tracking engagement and delivering insights, enabling pursuit of targeted outcomes
- Mentored team members on patterns, best practices, and front-end style; ensured creative concepts stayed within technically viable specifications

Hidden Level Games LLC, New York, NY*Co-founder & Lead Engineer, 10/2011–12/2017; Remote*

- Led engineering on a greenfield interactive programming game for Desktop (Windows/Mac) built in AS3 and Adobe AIR
- Designed and implemented CodePop, a custom domain-specific scripting language allowing real-time modification of game world and object properties — an early authoring framework for non-engineers
- Co-authored a Python backend (Flask, MongoDB), enabling users to save and share their creations
- Prototyped a WebSocket server for real-time multiplayer and collaborative experience development
- Partnered with the Anita Borg Foundation to host Beta coding camps; worked with educators to develop classroom curricula

EARLIER EXPERIENCE (Condensed)**Harrison and Star | Grey Group | Havas | Blue Telescope | Mercedes-Benz USA | Tremor Video***Freelance Senior Interactive Developer, 2010–2013 — Various Locations / Remote*

Engaged across multiple pharmaceutical, automotive, and media clients as a senior freelance engineer. Built interactive HTML5/CSS3/JavaScript content, rich media banner campaigns, iPad CRM/CLM applications, and networked multi-device applications. Created an app that leverages iFrames to load web-based content and the PostMessage API to update a shared shopping cart. Developed cookie-based auth systems and metrics tracking with session storage. Mentored junior developers, advised creative teams on technical feasibility, and delivered on-site support at conferences and events.

Visual Alchemy | FCB Health | LLNS | Erkel Associates*Freelance Developer / Lead Developer, 2009–2013 — New York, NY / Remote*

Led development of a content framework for direct sales platforms targeting web, DVD, and offline distribution. Architected scalable OOP-based frameworks, built XML and JSON based content management systems, introduced engineering best practices, and mentored and interviewed developers. Delivered interactive kiosks, microsites, and rich media applications for Pfizer, Boehringer-Ingelheim, and others.

IBM, Research Triangle Park, NC*Engineer/Scientist | Pre-Professional Co-op Engineer, 2004–2009*

Completed four consecutive summer co-op internships followed by a full-time role. Developed test applications in C# and ActionScript to validate hardware component stability. Work spanned schematic design, signal integrity, component testing, and networking.